1. Evaluate and Justify Dataset Choice

- **Source**: Combine the *Amazon Top 500 Bestsellers* dataset (Positive Class) with a sampled *Google Books* dataset (Negative Class).
- **Justification**: The Amazon dataset provides real-world examples of bestsellers. However, it lacks a negative class (non-bestsellers). To train a binary classifier, a large pool of general books is required for the negative class, which Google Books provides.
- **Quality Check**: Acknowledge early that the datasets are structurally different (Domain Shift), and Google Books has severe missingness in key columns (ratings, list_price), meaning we must rely on text-based metadata and category flags.

2. Plan the Analysis (EDA & Cleaning)

- **Target Leakage Prevention**:
  - **Amazon:** Drop post-publication metrics like `Rank`, `Reviews`, `Weeks on List`, `Amazon BSR`, and `Rating`. Keeping these would result in 100% trivial accuracy.
  - **Google:** Use Regex to filter out titles, categories, and publishers containing words like "bestseller", "award", or "prize" to prevent the model from learning keyword shortcuts.
- **Data Cleaning**: Address the extreme missingness in the Google dataset by dropping columns like `average_rating` and `page_count`.
- **Merge Attempt**: Try to join datasets via ISBN or normalized titles to enrich features (attempting to get page counts). *Expected outcome*: Recognize that the ISBNs are largely synthetic or mismatched, so abandon the merge and proceed with parallel feature engineering.

3. Feature Selection & Engineering Plan

- **Harmonization**: Create identical features for both datasets. Since numerical data (prices, ratings) is unreliable or missing, lean heavily on text-based features:
  - `title_word_count` & `title_char_count`
  - `author_word_count` & `publisher_word_count`
  - `is_fiction` (derived from Amazon Category and Google Categories)
  - `num_genres` (counting genres/sub-genres)

- **Multicollinearity Check**: Build a correlation matrix and drop features with high collinearity (e.g., drop `title_char_count` because it is perfectly correlated with `title_word_count`).
- **Class Balancing**: Sample 1,000 books from the Google dataset to pair with the 500 Amazon books, creating a 2:1 class ratio to avoid extreme imbalance.

4. Train Model Plan (From Scratch)

- **Algorithm**: Write a custom `ScratchLogisticRegression` class using NumPy.
- **Components**: Implement the Sigmoid function, Binary Cross-Entropy cost function, and Batch Gradient Descent. Include L1 (Lasso) and L2 (Ridge) regularization options.
- **Preprocessing**: Perform an 80/20 train-test split (stratified to maintain the class ratio). Use `StandardScaler` (fit only on the training set) to ensure features like word counts don't overpower binary features during gradient descent.
- **Hyperparameters**: Test learning rates (0.001, 0.01, 0.05, 0.1), iterations (500, 2000, 5000), and regularization strengths (0.01, 0.1) to check for stability and overfitting.

5. Interpret and Evaluate Model Plan

- **Metrics**: Evaluate the model using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Generate a Confusion Matrix and ROC curve.
- **Interpretation**: Analyze the feature coefficients to determine what drives a bestseller prediction (e.g., shorter titles, fiction status, number of genres).
- **Critique**: Note that while the metrics will likely be high (e.g., 88% accuracy, 0.92 AUC), this is likely inflated by the "Domain Shift" the model may be learning the difference between Amazon metadata and Google Books metadata rather than the true essence of a bestseller.

6. Write Report Plan

- Structure the final report with:
    - **Introduction**: Dataset choice and sourcing.
    - **EDA & Cleaning**: Missing value handling, leakage prevention, and visualizations (histograms, boxplots).
    - **Feature Engineering**: The harmonization process and multicollinearity checks.

- **Modeling**: Mathematics behind the custom logistic regression and hyperparameter tuning results.
- **Results**: Confusion matrices, ROC curves, and feature coefficients.
- **Limitations**: Explicitly discuss the dataset domain shift and its impact on model validity.